

A Country Nobody Lives In

84-355 Democracy's Data | Race in the 2020 census, from the nation to the block

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August 11, 2026

Everything here runs as given. **You will not write any code.** Your job is to read what comes out and say what it means.

Read `demographics-brief.Rmd` first. This lab runs the same argument as numbered steps with the code exposed. The chunks that *compute* are shown; the chunks that only *draw* are folded away, because their job is pictures, not arithmetic.

The claim under examination

“The United States is 57.8% White.”

True, on one of at least three defensible definitions, of a country in which almost nobody lives in a place that looks like that.

This lab holds the census fixed — one day, one enumeration, 331,449,281 people — and changes two things: **which definition** you count with, and **how big a box** you compute the average over.

Step 1 — The eight categories

```
c(us_population = FN("us_pop"),
  categories     = FN("n_categories"))
```

```
#> us_population    categories
#>    331449281          8
```

```
d <- data.frame(category = SHORT[CAT],
                 people   = as.numeric(nat[CAT]),
                 pct      = round(100 * as.numeric(nat[CAT]) / nat$total, 2))
d <- d[order(-d$people), ]
d$people <- format(d$people, big.mark = ",")
d
```

Category	People	Pct
White	191,697,647	57.84
Hispanic	62,080,044	18.73
Black	39,940,338	12.05
Asian	19,618,719	5.92
Two or More	13,548,983	4.09
AIAN	2,251,699	0.68
Some Other Race	1,689,833	0.51
NHPI	622,018	0.19

```
# the eight must partition the population exactly -- no remainder, no overlap
sum(as.numeric(nat[CAT])) == nat$total
```

```
#> [1] TRUE
```

These come from **Table P2** of the 2020 redistricting file, which cross-cuts the race question by the Hispanic-origin question and reports every resident exactly once. 331,449,281 people, 8 boxes, no remainder.

Two questions on the form, one table out. The **census** lab in this course is about those two questions and the write-in boxes behind them; this lab starts from the table they produce.

Step 2 — Nobody at the Census Bureau chose these boxes

```
o <- std[, c("issued", "citation", "categories", "question_form",
            "first_census")]
o$issued <- format(as.Date(o$issued), "%d %b %Y")
o
```

Issued	Citation	Categories	Question form	First census
12 May 1977	Statistical Policy Directive No. 15	5	separate collection preferred, not required	1980
30 Oct 1997	62 FR 58782	6	two separate questions preferred	2000
29 Mar 2024	89 FR 22182	7	a single combined question required	2030 (planned)

The categories are set by the **Office of Management and Budget** in Statistical Policy Directive No. 15, and they bind every federal agency that collects data on race. The directive has been issued three times, and each version changed what a person could be counted as:

```
YC <- paste0("c", c(1980, 1990, 2000, 2010, 2020, 2030))
o <- cts[cts$dimension == "race", c("category", YC)]
names(o) <- c("category", c(1980, 1990, 2000, 2010, 2020, "2030 (planned)"))
o
```

Category	1980	1990	2000	2010	2020	2030 (planned)
White	1	1	1	1	1	1
Black or African American	1	1	1	1	1	1
American Indian or Alaska Native	1	1	1	1	1	1
Asian or Pacific Islander	1	1	0	0	0	0
Asian	0	0	1	1	1	1
Native Hawaiian or Other Pacific Islander	0	0	1	1	1	0
Native Hawaiian or Pacific Islander	0	0	0	0	0	1
Some Other Race	1	1	1	1	1	0
Two or More Races	0	0	1	1	1	1
Middle Eastern or North African	0	0	0	0	0	1

Read the row for **Two or More Races**. It is empty until 2000, and not because nobody was of two races before 2000. Until the 1997 revision the form allowed one mark. **The category was created by a sentence in the Federal Register**, and the people it counted in 2000 had been in the country all along, in other boxes.

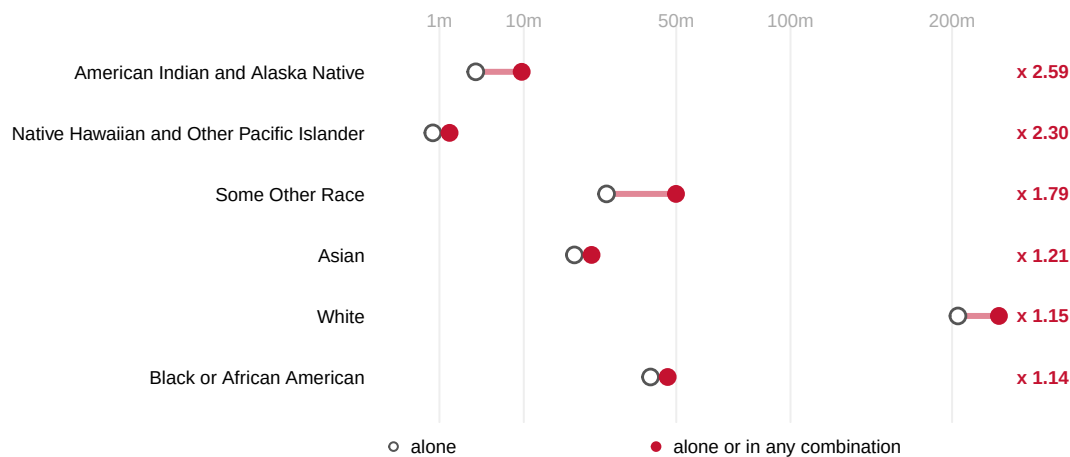
Read the row for **Middle Eastern or North African**. It appears only in the planned 2030 column. The 1977 directive defined White as covering origins in “Europe, North Africa, or the Middle East,” so until the 2024 revision the category absorbed them. Hold that fact: Step 9 lands on a county that contains the largest Arab-American concentration in the United States.

Before you run the next chunk, commit to a number. Step 3 counts each of the six races two ways — alone, and alone-or-in-any-combination. **Which category grows the most, in proportion, when the second mark is allowed, and by what factor?** Write down a name and a multiplier before you run it.

Step 3 — Alone, or in any combination

```
o <- data.frame(
  race = RLAB[RACE6],
  alone = vapply(RACE6, function(r) FN(paste0("us_", r, "_alone")), 0),
  any = vapply(RACE6, function(r) FN(paste0("us_", r, "_any")), 0))
o$difference <- o$any - o$alone
o$ratio <- round(o$any / o$alone, 2)
o[order(-o$ratio), ]
```

Race	Alone	Any	Difference	Ratio
American Indian and Alaska Native	3727135	9666058	5938923	2.59
Native Hawaiian and Other Pacific Islander	689966	1586463	896497	2.30
Some Other Race	27915715	49902536	21986821	1.79
Asian	19886049	24000998	4114949	1.21
White	204277273	235411507	31134234	1.15
Black or African American	41104200	46936733	5832533	1.14



American Indian and Alaska Native goes from 3,727,135 to 9,666,058, a factor of 2.59. White moves by the largest absolute amount — 31,134,234 people — and by the smallest proportion, 1.15.

Almost everyone predicts Black or White, because those are the categories the first two steps put in your head. The rule is the opposite: **the smaller the category, the more the second mark matters.** And the small categories are exactly the ones that voting-rights law, tribal funding formulas and civil-rights enforcement are written about.

Step 4 — Two censuses, and why they do not compare

```
K <- c("white_alone", "white_any", "black_alone", "black_any",
      "aian_any", "sor_any", "two_plus")
data.frame(
  measure = K,
  y2010 = vapply(K, function(k) FN(paste0("d10_", k)), 0),
  y2020 = vapply(K, function(k) FN(paste0("d20_", k)), 0),
  pct_change = vapply(K, function(k) FV(paste0("chg_", k)), ""))
```

Measure	Y2010	Y2020	Pct change
white_alone	223553265	204277273	-8.6
white_any	231040398	235411507	+1.9
black_alone	38929319	41104200	+5.6
black_any	42020743	46936733	+11.7
aian_any	5220579	9666058	+85.2
sor_any	21748084	49902536	+129.5
two_plus	9009073	33848943	+275.7

White alone falls 8.6% and White in any combination rises 1.9%. Two or More Races rises 275.7%.

Do not read those as demographic change. Between the two censuses the Bureau added dedicated write-in areas under “White” and under “Black or African American,” raised the characters captured per write-in from 30 to 200, and began assigning up to six detailed codes per response instead of two. Of the jump in Some Other Race the Bureau’s own analysts wrote that it “could be attributed to a number of factors, including demographic change since 2010. But we expect they were largely due to the improvements to the design of the two separate questions for race and ethnicity, data processing and coding.”

Notice how hard this failure is. **2010 and 2020 are the easy pair**: same OMB standard, same categories, same file format, same fields in the same order. Everything that could be held constant was, and the comparison still cannot be trusted. A comparison crossing 1997 has no chance at all.

Step 5 — Regions and divisions

```
o <- reg[, c("kind", "name", "total", CAT[1:4], "entropy")]
o$diversity <- round(100 * o$entropy / log(8), 1)
o$entropy <- NULL
o[order(o$kind, -o$total), ]
```

Kind	Name	Total	Hispanic	Nh white	Nh black	Nh asian	Diversity
division	South Atlantic	66089734	10147394	36365960	13714620	2653826	60.1
division	Pacific	53669422	17416369	22405688	2536722	7468661	67.4
division	East North Central	47368533	4424621	33477559	5595258	1717090	48.3
division	Middle Atlantic	42492943	7000222	24968705	5282142	3365924	59.5
division	West South Central	40774139	12493044	18652037	5630258	1787717	64.2
division	Mountain	24919150	6193275	15062974	930550	863077	55.8
division	West North Central	21616921	1554165	16604993	1516295	668601	43.2
division	East South Central	19402234	1056308	13375438	3808696	316368	46.3
division	New England	15116205	1794646	10784293	925797	777455	48.7
region	South	126266107	23696746	68393435	23153574	4757911	60.9
region	West	78588572	23609644	37468662	3467272	8331738	65.0
region	Midwest	68985454	5978786	50082552	7111553	2385691	47.0
region	Northeast	57609148	8794868	35752998	6207939	4143379	57.1

Composition varies enormously by region. But the smallest of the four regions holds 57,609,148 people, and an average over a container that size describes nowhere in particular.

Step 6 — States

```
sta$diversity <- round(100 * sta$entropy / log(8), 1)
sta$white_pct <- round(100 * sta$nh_white / sta$total, 1)

head(sta[order(-sta$diversity), c("name", "total", "white_pct", "largest",
                                "diversity")], 6)
```

Name	Total	White pct	Largest	Diversity
Hawaii	1455271	21.6	Asian	75.6
Nevada	3104614	45.9	White	67.7
Alaska	733391	57.5	White	66.4
California	39538223	34.7	Hispanic	66.3
New York	20201249	52.5	White	64.2
Maryland	6177224	47.2	White	64.0

```
head(sta[order(sta$diversity), c("name", "total", "white_pct", "largest",
                                "diversity")], 6)
```

Name	Total	White pct	Largest	Diversity
Maine	1362359	90.2	White	22.8
West Virginia	1793716	89.1	White	23.9
Vermont	643077	89.1	White	24.3
New Hampshire	1377529	87.2	White	27.5
Montana	1084225	83.1	White	33.4
Iowa	3190369	82.7	White	34.4

```
# the states whose largest group is not White
sta[sta$largest != "White", c("name", "largest", "diversity")]
```

Name	Largest	Diversity
California	Hispanic	66.3
District of Columbia	Black	62.7
Hawaii	Asian	75.6
New Mexico	Hispanic	58.1

Hawaii scores 75.6 and Maine scores 22.8. Only 4 states have a largest group that is not White.

Step 7 — The measure, and the control

The **diversity score** used from here on is the multigroup entropy index, expressed as a percentage of its maximum. For a unit with shares p_r over the eight categories, $E = \sum_r p_r \ln(1/p_r)$; it is 0 when everyone is in one category and $\ln 8$ when the eight are perfectly even.

```
c(E_united_states = FN("E_us"),
  E_maximum       = FN("E_max"),
  diversity_score = FN("div_nation"))
```

```
#> E_united_states      E_maximum diversity_score
#>      1.2560          2.0794          60.4000
```

Entropy rather than the more famous dissimilarity index, for one reason: dissimilarity takes exactly **two** groups. With eight categories that means either 28 numbers or a decision that one contrast — nearly always Black against White — is *the* contrast. That decision is a claim about the country, and the measure should not make it silently.

The measure has a real problem, and Step 8 controls for it. **Smaller units always look more segregated, even if people were sprinkled at random**, because a unit of twelve people cannot be evenly mixed across eight categories. So the build script also computes a **null**: the same population shuffled at random into units of the identical sizes.

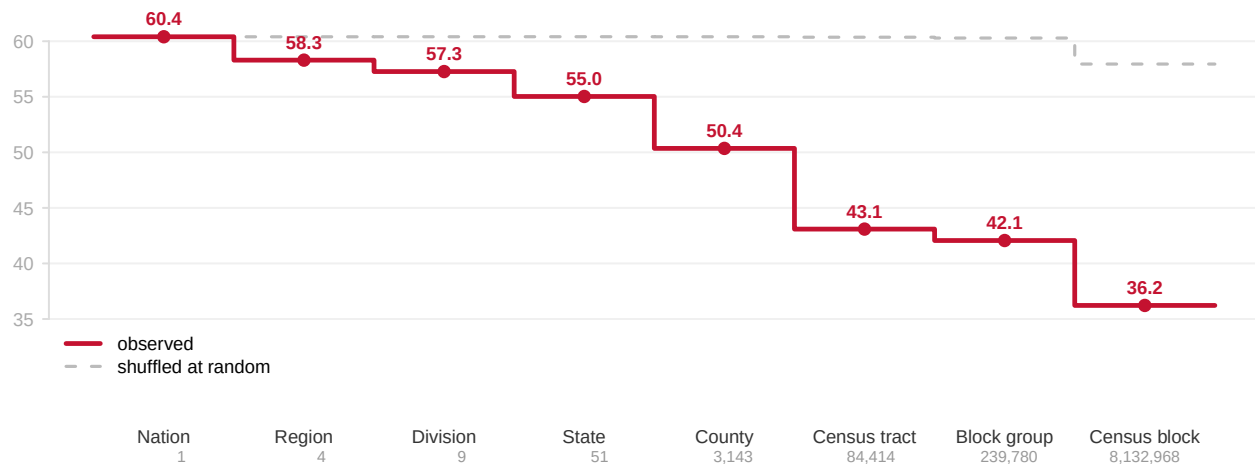
```
o <- lad[, c("level", "units", "diversity", "H", "H0")]
o$H <- round(o$H, 4); o$H0 <- round(o$H0, 4)
o
```

Level	Units	Diversity	H	H0
Nation	1	60.40302	0.0000	0.0000
Region	4	58.29831	0.0348	0.0000
Division	9	57.27177	0.0518	0.0000
State	51	55.02968	0.0890	-0.0001
County	3143	50.35530	0.1663	0.0000
Census tract	84414	43.09214	0.2866	0.0007
Block group	239780	42.06346	0.3036	0.0019
Census block	8132968	36.22271	0.4003	0.0407

H is the share of national diversity missing from the average person's unit. H0 is the same number for the shuffled population. **Read them as a pair.**

Step 8 — The ladder

Diversity score of the unit the average American lives in



```
c(nation = FN("div_nation"),
  state   = FN("div_state"),
  county  = FN("div_county"),
  tract   = FN("div_censustract"),
  block   = FN("div_censusblock"))
```

```
#> nation state county tract block
#> 60.4 55.0 50.4 43.1 36.2
```

```
# the same climb, with the null beside it
o <- lad[, c("level", "H", "H0", "H_net")]
o[, 2:4] <- round(o[, 2:4], 4)
o
```

Level	H	H0	H net
Nation	0.0000	0.0000	0.0000
Region	0.0348	0.0000	0.0348
Division	0.0518	0.0000	0.0519
State	0.0890	-0.0001	0.0890
County	0.1663	0.0000	0.1663
Census tract	0.2866	0.0007	0.2859
Block group	0.3036	0.0019	0.3017
Census block	0.4003	0.0407	0.3596

The country scores 60.4; the average American's census tract scores 43.1. The dashed line is flat at the national value all the way down to block groups — so the fall is **not** an artefact of using smaller boxes. Only at the block rung, where a unit holds a few dozen people, does randomness start producing apparent segregation by itself, and even there most of the effect survives the control.

Without any index at all:

```
c(pct_in_tract_at_least_80_pct_one_group = FN("pop_tract_80plus"),
  pct_in_tract_at_least_90_pct_one_group = FN("pop_tract_90plus"),
  pct_in_tract_whose_largest_group_under_half = FN("pop_tract_under50"))
```

```
#>      pct_in_tract_at_least_80_pct_one_group
#>                                     33.9
#>      pct_in_tract_at_least_90_pct_one_group
#>                                     13.1
#> pct_in_tract_whose_largest_group_under_half
#>                                     17.1
```

33.9% of Americans live in a census tract where at least four residents in five are in the same category.

Compare this with `areal-units`, which runs a similar ladder to make the *opposite* point: that aggregation destroys tails and flips correlations. That lab is about the measuring instrument. Step 8 is what is left when you subtract the instrument.

Step 9 — One county

```
c(wayne_population = FN("wayne_pop"),
  tracts           = FN("wayne_tracts"),
  block_groups     = FN("wayne_bg"),
  blocks           = FN("wayne_blocks"))
```

```
#> wayne_population      tracts    block_groups      blocks
#>      1793561           627         1507         32022
```

```
# the county row: no majority group at all
round(100 * as.numeric(cou[cou$geoid == "26163", CAT]) /
      cou$total[cou$geoid == "26163"], 1) -> v
setNames(v, SHORT[CAT])
```

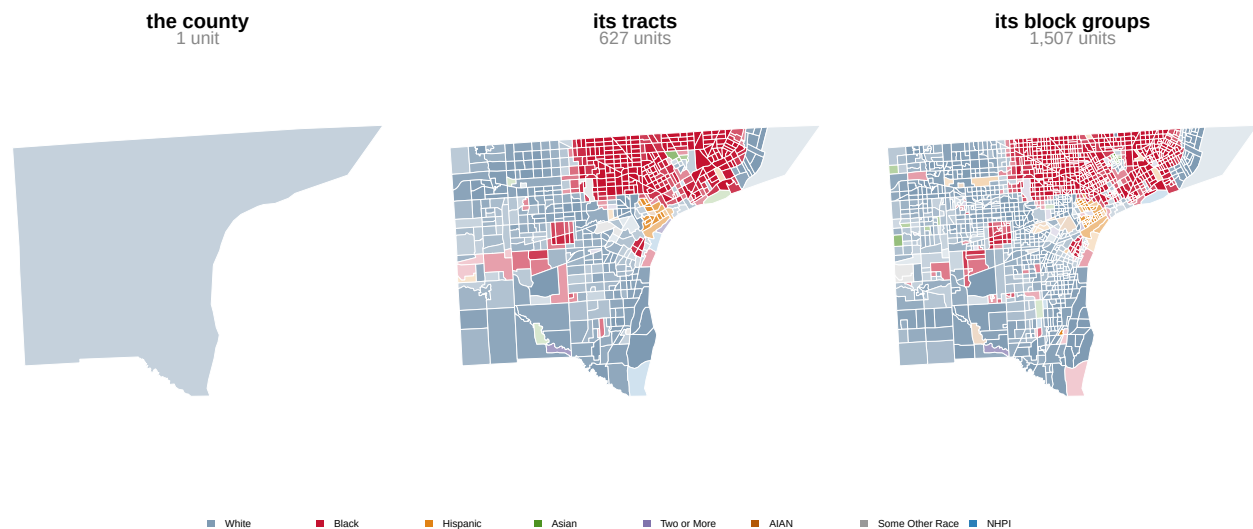


```
#>      Hispanic      White      Black      Asian      Two or More
#>      6.6      47.8      37.3      3.6      4.0
#>      AIAN Some Other Race      NHPI
#>      0.2      0.4      0.0
```

```
c(diversity_score = FN("wayne_div"),
  rank_of_3143_counties = as.integer(rank(-cou$entropy)[cou$geoid == "26163"]))
```

```
#>      diversity_score rank_of_3143_counties
#>      57.2      196.0
```

Wayne County, Michigan contains Detroit, has no majority group, and sits in the most diverse 6.2% of American counties. From the county row alone you would call it integrated.



```
table(largest_group_in_tract = wtr$largest)
```

```
#> largest_group_in_tract
#>      AIAN      Asian      Black      Hispanic      NHPI
#>      1      6      279      21      3
#> Some Other Race      Two or More      White
#>      5      5      307
```

```
c(median_tract_pct_one_group = FN("wayne_tract_median_largest"),
  tracts_at_least_80_pct_black = FN("wayne_tracts_80_black"),
  tracts_at_least_80_pct_white = FN("wayne_tracts_80_white"),
  pct_of_county_in_those_tracts = FN("wayne_pop_80_pct"),
  tracts_at_least_90_pct_one_group = FN("wayne_tracts_90"),
  pct_of_county_in_those = FN("wayne_pop_90_pct"))
```

```
#>      median_tract_pct_one_group      tracts_at_least_80_pct_black
#>      81.6      201.0
#>      tracts_at_least_80_pct_white      pct_of_county_in_those_tracts
#>      125.0      51.8
#>      tracts_at_least_90_pct_one_group      pct_of_county_in_those
#>      156.0      22.9
```

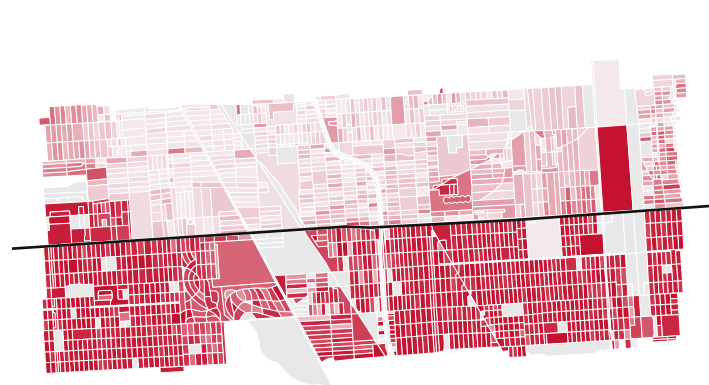
The median Wayne County tract is 81.6% a single category, and 51.8% of the county's residents live in a tract that is at least 80% one group. The county row and the tract rows answer different questions, and only the tract rows are about anybody's street.

Step 10 — One street

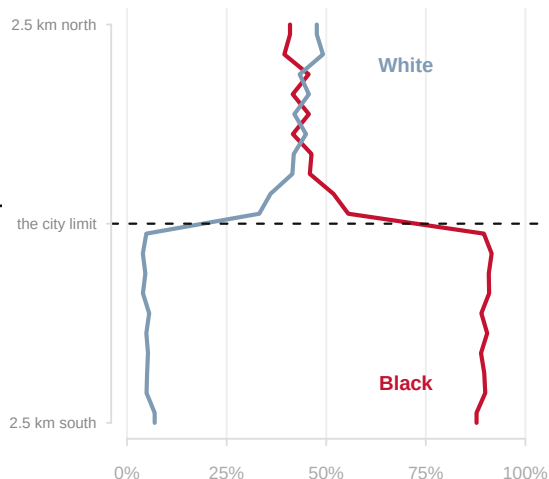
The boundary you can see in the middle panel is Eight Mile Road, the northern city limit of Detroit. Along this stretch it is **also the county line**, which means no county-level statistic can show it.

blocks along Eight Mile Road

shading = share Black; black line = Detroit city limit



composition, by distance from it



```
c(blocks_in_the_profile = FN("profile_blocks"),
  people_in_the_profile = FN("profile_pop"),
  pct_black_south_of_the_line = FN("prof_south_black"),
  pct_black_north_of_the_line = FN("prof_north_black"),
  pct_black_last_500m_inside = FN("prof_in500_black"),
  pct_black_first_500m_outside = FN("prof_out500_black"),
  the_step_in_points = FN("prof_jump_black"))
```

```
#>      blocks_in_the_profile      people_in_the_profile
#>                5445.0                299535.0
#> pct_black_south_of_the_line pct_black_north_of_the_line
#>                89.9                45.2
#> pct_black_last_500m_inside pct_black_first_500m_outside
#>                90.7                53.5
#>      the_step_in_points
#>                37.2
```

Across one road the Black share falls 37.2 points.

Look at the *shape* rather than the levels. It is flat, then a step, then flat again. Composition does not fade with distance from Detroit the way something diffusing would; it changes at the boundary and stops. Whatever produced this was operating on the line itself — on which side of a municipal border a house sits — and not on distance from a city centre.

Step 11 — What you can now say

You can say what the United States looked like in 2020, in eight categories, and give the numbers.

You can say that “the Black population” and “the White population” are quantities with at least two correct values each, name the gap (2.59× for American Indian and Alaska Native, 1.15× for White), and explain which OMB revision created the ambiguity and when.

You can say that the 2010-to-2020 change in the multiracial and Some Other Race counts is largely a measurement change, and cite the Bureau saying so.

You can say that the diversity of the unit the average American lives in falls from 60.4 to 43.1 between the nation and the census tract, **and** that a random-shuffle control rules out the obvious objection, because the null stays flat at 0.0007 through the tract rung.

You cannot say why anyone lives where they live. Nothing here has a mechanism in it.

You cannot say that any of these numbers is definition-free. Change the categories — which OMB has now done — and every figure moves.

Your turn

Change **one** thing, re-run, and write about what changed. Pick one.

Option A — A different county. `data/counties.csv` holds all 3,143 counties with the eight categories and a diversity score. Find one whose county-level score is high and say what you would need in order to know whether it is actually integrated. Then find a county whose score is *low* and ask whether the county row is misleading in the other direction.

Option B — A different definition. Step 3 gives every race two counts. Recompute the state table in Step 6 using `black_any` instead of `nh_black`, and report which states move most. Which definition would you defend in a voting-rights case, and why?

Option C — A different threshold. Step 8 reports the share of Americans in a tract that is at least 80% one group. `data/tract_largest.csv` has the whole distribution in ten-point bins. At what threshold does the statistic stop sounding alarming, and what does the fact that you can choose tell you about the statistic?

Option D — The 2024 categories. The revised Directive 15 has 7 categories rather than eight, and moves Hispanic origin into the race list. Using `data/categories.csv`, work out which of this lab’s numbers would change, which would not, and which are impossible to predict without new data.

What this lab cannot tell you

- **Whether the categories are the right ones.** They are administrative categories set by OMB and implemented by the Census Bureau. Every figure here is downstream of them.
- **How the country has changed.** See Step 4. The data will not carry a clean time series and the honest thing is to say so.
- **Anything exact about a single block.** The 2020 census applied a disclosure-avoidance system based on differential privacy. State totals were held invariant; **block-level counts were not**, and the noise is proportionally largest where units are smallest.
- **Anything about voters.** These are residents, including children and non-citizens. Voting-age population is in the same file and gives different numbers.

- **Whether Wayne County is typical.** It was chosen because its county row is misleading in an instructive direction. Step 8 is the national check that stops the county example doing work it cannot do.

On the discussion board

By **Friday, 11:59 p.m.**

- Step 3 shows the same population producing two correct counts. If a news story reports “the Black population of the United States,” what is the one question you now have to ask before believing the number?
- Step 4 shows a decade of apparent demographic change that the Census Bureau attributes largely to a better form. Whose fault is a headline like “the White population declined for the first time in history,” and what should the Bureau have done differently — if anything?
- Step 8 shows the country scoring 60.4 and the average American’s neighbourhood scoring 43.1. Which of those two numbers should appear in a news story about American diversity, and does your answer change if the story is about school assignment rather than national identity?

Where the data came from

- **Race and Hispanic origin at every level** — U.S. Census Bureau, *2020 Census Redistricting Data (P.L. 94-171) Summary File*, legacy format, Tables P1 and P2. The national roll-up plus all 51 state files, giving the nation, 4 regions, 9 divisions, 51 states, 3,143 counties, 84,414 tracts, 239,780 block groups and 8,132,968 blocks. Fetched 2026-08-11. Keyless; **this lab uses no Census API and contains no API key.** https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File--PL_94-171/
- **The 2010 comparison** — the same file for 2010, national roll-up only. https://www2.census.gov/census_2010/redistricting_file--pl_94-171/National/
- **Boundaries** — Census TIGER/Line 2020: states, and for Michigan the tracts, block groups, blocks and place boundaries. <https://www2.census.gov/geo/tiger/TIGER2020/>
- **The categories** — Office of Management and Budget, Statistical Policy Directive No. 15 (1977), 62 FR 58782 (1997), and 89 FR 22182 (2024).
- **The measure** — Theil, H. (1972), *Statistical Decomposition Analysis*; Reardon, S. F. and Firebaugh, G. (2002), “Measures of Multigroup Segregation,” *Sociological Methodology* 32(1).

The build is `data/build-data.R`. It records its source URLs, fetch date and row counts, checks that the eight categories partition the population at every level of every state, and stops if a state’s blocks do not sum to the state.